ARMY 2040: WORLD-CLASS. MULTI-MISSION READY. CROSS-DOMAIN CAPABLE.

ELECTRONIC COUNTER-COUNTERMEASURES

ECCM · Electronic Protection for the soldier

SOAC

Sgt John Venhur G Esver (SC) PA
Squad leader, ES Squad, 1st ES Pltn, Electronic Warfare Company (P), SIMBn, SR, PA

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HONOR. PATRIOTISM. DUTY.

Lesson Overview and Objectives

TERMINAL LEARNING OBJECTIVE

Upon completion, students will explain the purpose of ECCM / Electronic Protection and apply basic measures that keep friendly systems working and hidden under enemy electronic attack.

By the end of this lecture you can:

ELO 1

Define ECCM / EP and its link to ECM

ELO 2

Apply radio discipline, masking and EMCON

ELO 3

Recognize, react to and report jamming

ELO 4

Use PACE and protect GPS when systems fail

DOCTRINAL BASIS

PAM 3-0702-01 Electronic Warfare Operations (Sep 2026)

Phones on silent · Questions anytime · About 45 min

Ice Breaker: Jammed or Broken?



"...SSSHHHHKKKKRRRR..."

Mid-mission, 0215H

ICE BREAKER

Your radio suddenly fills with loud static in the middle of a mission.

List three possible causes before you blame the enemy.

1. _____
2. _____
3. _____

Ice Breaker: The Answer



Enemy jamming

Deliberate noise or false signals on your frequency.



Friendly interference

Our own radios, generators or C-RCIED nearby.



Equipment fault

Loose antenna, bad cable, low battery, wrong setting.



Terrain and distance

Hills, buildings or range weaken the wanted signal.

A flat tyre isn't always a sniper. Diagnose before you blame the enemy.

The Threat Is Already Here



Schematic. Pin positions approximate.

1 **Feb 2024 · West Philippine Sea**

PN: increased jamming affected ships' and land-based comms, mostly cellphones.

2 **2016 / 2024 · Airport approaches**

Pilots lose GPS on approach. CAAP now requires GNSS interference reports.

! **2022–2025 · Ukraine**

Command posts and drone teams found by their emissions, then hit by fires.

If we can't protect our signals, we can't fight.

ECM vs ECCM: The Duel

ENEMY

Spot-jams our channel

US

Radios hop frequency

ENEMY

Switches to barrage

US

Lower power, mask with terrain, switch to PACE

ECM (now Electronic Attack): attacks our use of the spectrum.

ECCM (now Electronic Protection): keeps our systems working anyway.

Lock and lockpick: every move invites a counter-move.

ECCM = Electronic Protection

PAM 3-0702-01: *"Units implement EP to safeguard friendly personnel, facilities, and equipment from the effects of electromagnetic spectrum emissions, including friendly, adversary, or natural electromagnetic interference."*

THE FIVE EP MEASURES IN PAM 3-0702-01 (para 0110)

- **Hardening**
- **Encryption**
- **Frequency hopping**
- **Emission control**
- **Spectrum management**

Every unit does EP, not just EW units.

Three Questions the Enemy Asks

Can I FIND you?

Direction finding locates your transmitter.

OUR EP ANSWER

- Emission control
- Low power, short transmissions
- Terrain masking

Can I JAM you?

Electronic attack drowns out your signal.

OUR EP ANSWER

- Frequency hopping
- Directional antennas
- PACE alternate means

Can I READ you?

Intercept exploits what you say.

OUR EP ANSWER

- Encryption (secure mode)
- Brevity codes
- Say less, less often

Be hard to find, hard to jam, and hard to read.

Your First Defence: Radio Discipline

Lowest power that works

Less power, less range for enemy DF.

Short transmissions

Pre-plan the message; use brevity codes.

Transmit only when needed

No chatter, no radio checks for comfort.

Directional antennas

Point the energy at your station, not the enemy.

Smart antenna placement

Off skylines; use offset or remote antennas.

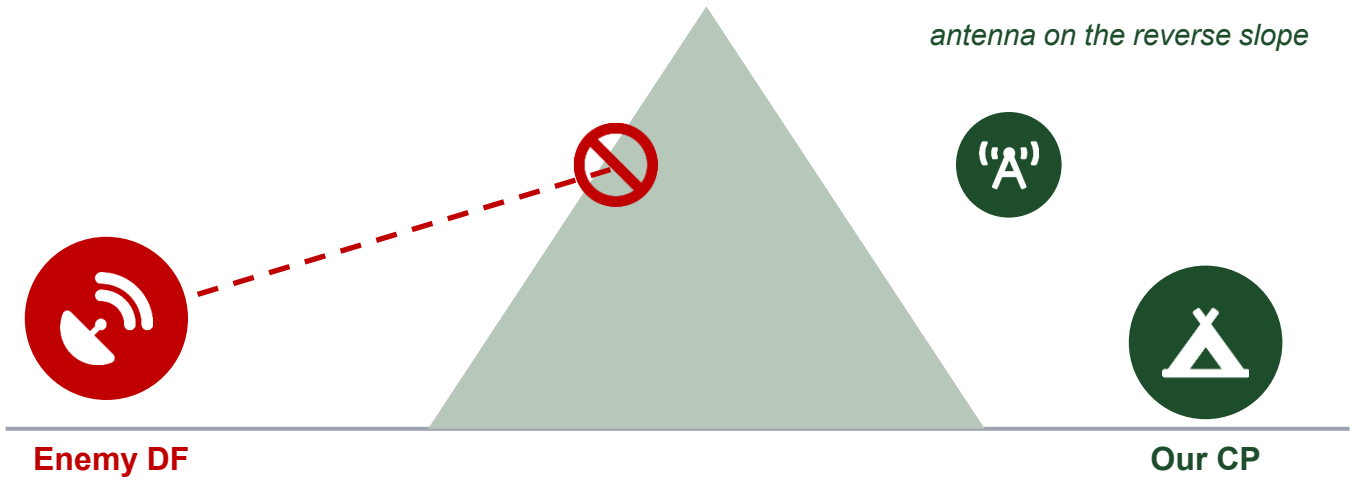
Vary your routine

No fixed report times or patterns.

PAM's campfire: the smaller the fire, the harder you are to see.

Hide Behind the Hill: Terrain Masking

Terrain between us and the enemy blocks his DF and jamming.

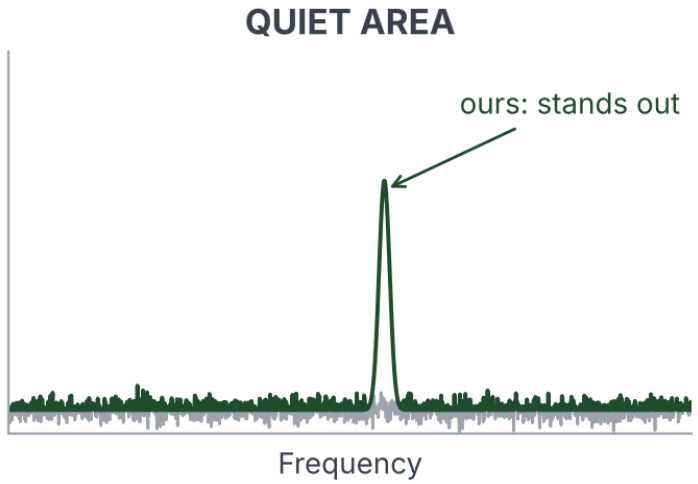


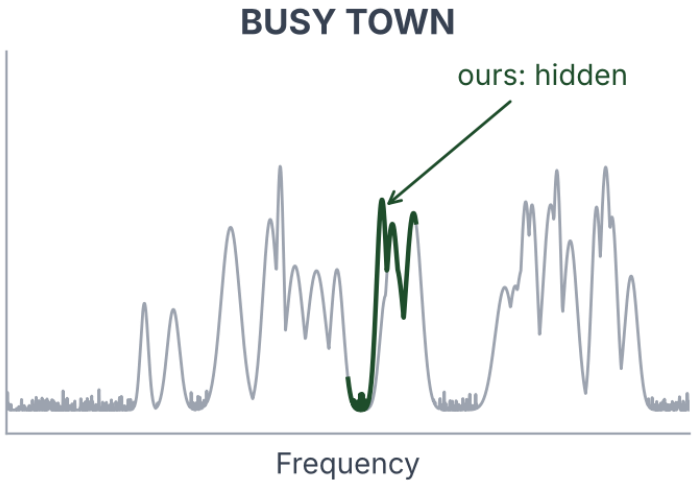
- Put terrain between your antenna and the enemy.
- Avoid hilltops for CPs: easy to DF, easy to jam.
- Use the reverse slope or the side of a building.
- Coordinate site and route choice with S3 and S6.

Camouflage nets on the sides and back of directional antennas block stray radiation.

Stand behind the wall, not on top of it.

Blend In: Hide a Needle Among Needles





- Use the spectrum the way the people around you use it.
- Odd military signals stand out in a quiet area.
- Avoid predictable report times and patterns.

Ukraine: busy commercial networks near Kyiv helped mask military signals.

A needle in a haystack can be found with a magnet. Hide it among other needles.

EMCON Levels 1 to 5

More restrictive →

EMCON 1
GREEN
No hostile activity. Normal operations, encryption on.

EMCON 2
YELLOW
Risk of attack. Limit transmissions; watch power levels.

FIELD BASELINE

EMCON 3
AMBER
Risk identified. Encrypt, hop, directional antennas; unencrypted systems off.

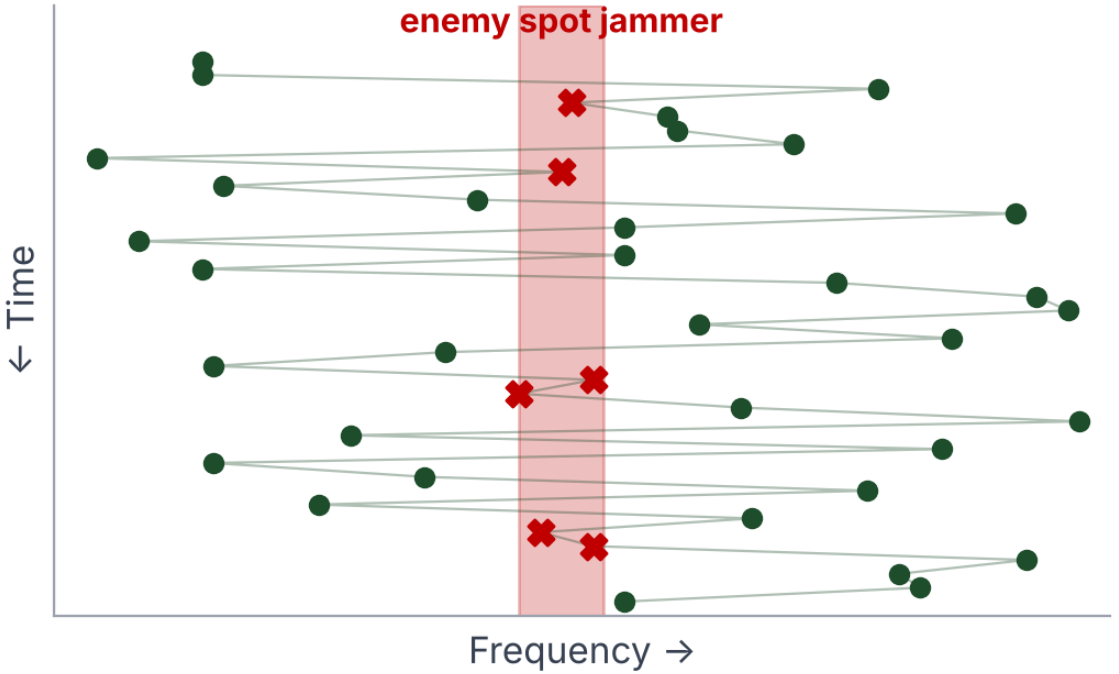
EMCON 4
RED
Attack has occurred. Non-essential emitters off; use alternates.

EMCON 5
BLACK
Attacks under way. Most restrictive; isolate compromised systems.

Rehearse moving between levels in every exercise, not just on paper.

FM 3-12 (2025) Table 3-1; PAM 3-0702-01 para 0110d

Frequency Hopping and Spread Spectrum



Only 6 of 40 hops land in the jammer's channel.

- The radio jumps channels many times per second in a pattern only friendly radios know.
- A spot jammer hits only a few hops; error correction recovers the rest.
- Spread spectrum spreads the signal so it is harder to detect and jam.

OUR RADIOS
PA tactical radios (Harris radios, Combat Net Radio System) are required to have hopping and encryption. Use them in hop mode.

PAM: soldiers who keep moving after every throw are hard for the sniper to hit.

Encryption ≠ Anti-Jam

ENCRYPTION

Protects WHAT you say

- Enemy hears gibberish, not your plan
- Does NOT stop jamming
- Does NOT hide your location

ANTI-JAM

Protects WHETHER it gets through

- Hopping, power, antennas, terrain
- Keeps the link alive under attack
- Low power and masking also hide you

You need both. Your tactical radio gives you both, if you set it to HOP and SECURE.

Am I Being Jammed? Recognize

SIGNS

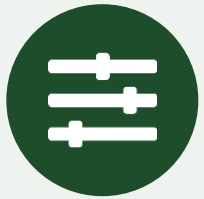
- Steady noise, buzzing or tones
- Squelch opens with no one talking
- Broken voice; data won't go through
- Every station drops at once
- GPS position jumps, freezes or is wrong

THE QUICK TEST

1. **Disconnect the antenna.**
   Noise drops? It is coming from outside: jamming or interference.
2. **Noise stays?**
   Suspect your own radio, cables or power.
3. **Move a short distance or turn the antenna.**
   Big change? Terrain or direction is involved.

Same symptoms, different causes: test before you report.

Am I Being Jammed? React

STEP 1	STEP 2	STEP 3	STEP 4
			
Keep operating	**Check equipment**	**Adjust**	**Report**
Never announce in the clear that you are jammed. Don't show the enemy he succeeded.	Antenna, cables, battery, settings and fills.	Raise power briefly, turn or change the antenna, move behind cover, go to the alternate frequency or means.	Tell your higher HQ by an alternate means if needed.

Don't let the enemy see you flinch.

Report It: Interference Report

INTERFERENCE REPORT (generic training format)	
1 WHO	Unit and call sign reporting
2 WHEN	Date-time group it started (and stopped)
3 WHERE	Your grid; bearing to source if known
4 FREQUENCY / NET	Affected frequency, net or system (e.g., GPS)
5 WHAT	Type: noise, tones, voice, false signals
6 EFFECT	What the mission lost: voice, data, position
7 ACTIONS	What you did and the result

- Send it through the chain to the S6 and the EW cell.
- Use an alternate means if the net is jammed.
- Never send it in the clear on the jammed net.
- Your report feeds ES, the EOB and the commander's response.

No PA standard format exists yet. Use this until one is published.

One soldier's report can reveal an enemy jammer to the whole force.

PACE: When the Radio Fails

P PRIMARY

A ALTERNATE

C CONTINGENCY

E EMERGENCY

EXAMPLE PLAN

P Tactical radio (CNRS / Harris) in hop and secure mode

A HF radio or another frequency or net

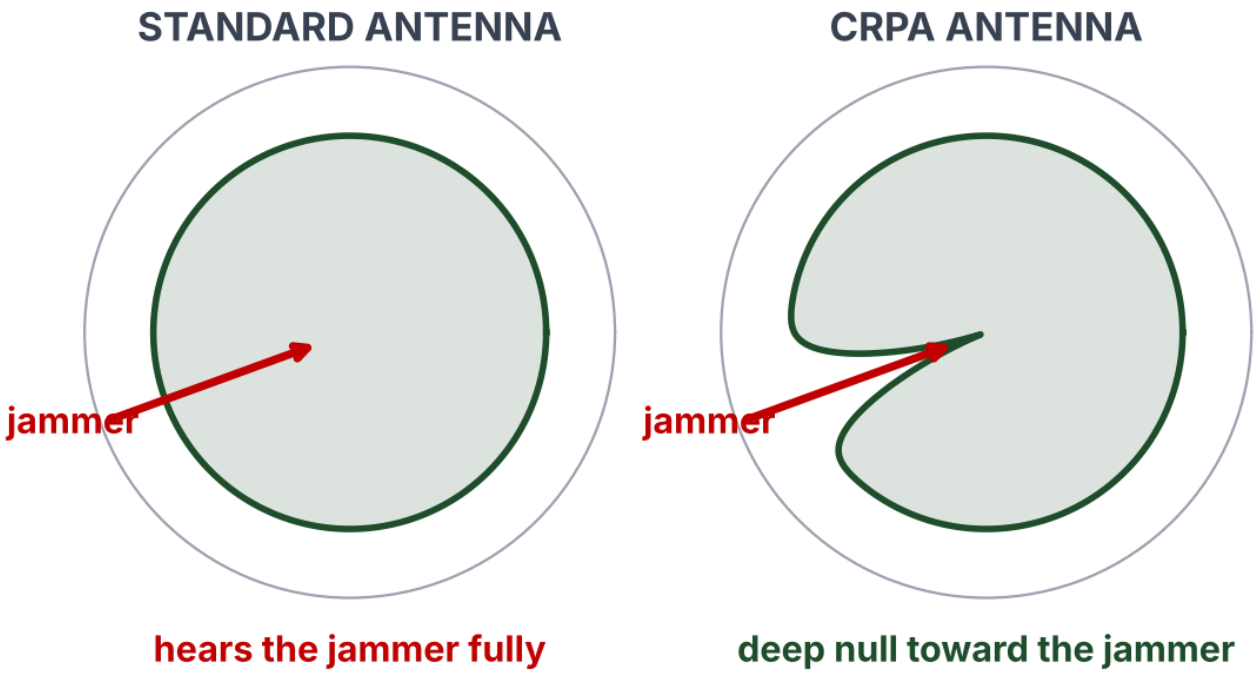
C Field wire, or a messenger vehicle

E Runner, or visual and pyrotechnic signals

Rehearse the switch. Know the trigger.

Spare tyre, jack, tow truck, walk: always have the next option ready.

Protecting GPS / PNT



Jamming = no fix. Spoofing = a wrong fix that looks right.

- Cross-check GPS with map, compass and terrain.
- Spoofing red flags: position jumps, wrong time, speed while stationary.
- CRPA antennas null out jammers.
- Report GPS interference like any other.

Your compass can't be jammed. Keep the land-navigation skills sharp.

Hardening and Compatibility

HARDENING

Protect equipment from damaging energy and jamming (PAM 0110a).

SHIELDING | FILTERING | GROUNDING

BONDING | ATTENUATING | BLANKING

Check: cable shielding, connectors, grounding on every setup.

COMPATIBILITY

Keep our own emitters from jamming each other (spectrum management, PAM 0110e).

- Separate antennas; don't co-site on the same frequencies
- C-RCIED jammers must not blind the convoy net
- Use only assigned frequencies from the S6

PAM: hardening puts soldiers in an armoured, waterproof shelter instead of a field uniform.

Sensor ECCM: Radar Example

CONCEPT EXPLAINED WITH RADAR AND AIR ASSETS

Frequency agility
The radar changes frequency pulse to pulse, like a hopping radio.

Pulse compression
A long, coded pulse is squeezed on receive for range detail and anti-jam margin.

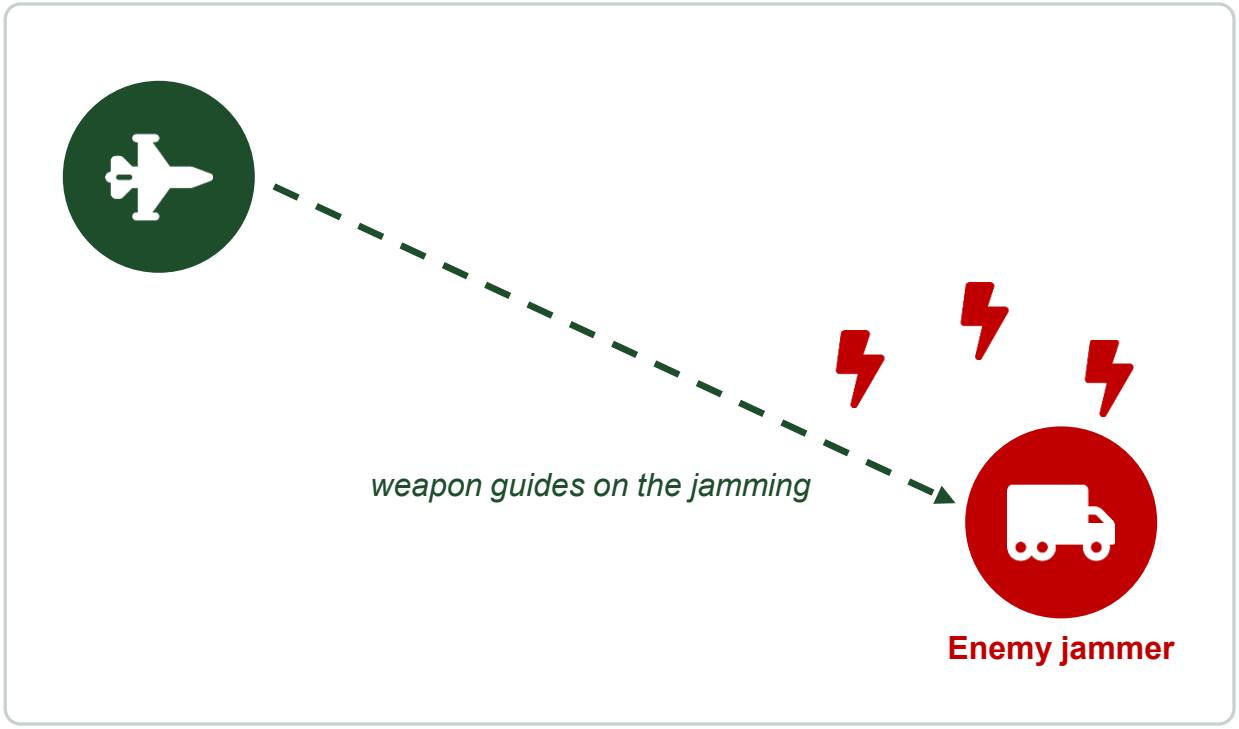
Sidelobe blanking
A second antenna spots jamming arriving from the side, and the radar ignores it.

Low-probability-of-intercept
Low-power, wideband waveforms that are hard for ES to detect.

Same principle as our radios: be unpredictable, reject noise, and be hard to detect.

Home-on-Jam: The Jammer Becomes a Target

CONCEPT EXPLAINED WITH AN AIR ASSET



- Some weapons can switch to guiding on a jamming source.
- Lesson for us: any strong emitter, jammer or radio, can be located and struck.
- Our own EA teams face the same risk (EA lecture).

Shouting in the dark tells everyone where you are.

Ukraine Lessons for EP



Static emissions get hit

Command posts were high-payoff targets because their emissions were stationary and constant.



DF leads to fires

Drone control stations were located by direction finding and engaged by indirect fire.



Blending works

Busy civilian networks helped mask military signals.

Move, mask, minimise: emit like your life depends on it, because it does.

Key Takeaways

1

EP is every soldier's job

Not just EW units. PAM 3-0702-01 and FM 3-12 require it.

2

Hard to find, jam and read

Radio discipline, masking, EMCON, hopping and encryption, together.

3

Diagnose, react, report

Then fall back on PACE when the radio fails.

"Victory in Electronic Warfare belongs not only to those who disrupt, but to those who endure disruption."

Check on Learning

1. According to PAM 3-0702-01, what does Electronic Protection safeguard us from?
2. Why doesn't encryption alone protect you from jamming?
3. Which EMCON level is the recommended field baseline, and what does it require?
4. Your radio fills with noise. What is the quick test to tell jamming from a fault?
5. Give an example PACE plan for your section.



END OF PRESENTATION

Electronic Counter-Countermeasures (ECCM)

Questions?